

* Simultaneous Equation -

Powers:

$2^2 = 4$	$3^2 = 9$	$4^2 = 16$	$5^2 = 25$
$2^3 = 8$	$3^3 = 27$	$4^3 = 64$	$5^3 = 125$
$2^4 = 16$	$3^4 = 81$	$4^4 = 256$	$5^4 = 625$
$2^5 = 32$	$3^5 = 243$	$4^5 = 1024$	$5^5 = 3125$

- 1 eqⁿ 1 Unknown $2x - 13 + 91 = x + 120$ ✓
- 1 eqⁿ 2 Unknowns $x + y = 10$ ✗
- 2 eqⁿ 2 Unknowns $\begin{cases} x + y = 10 \\ x - y = 9 \end{cases}$ ✓
- 2 eqⁿ 3 Unknowns $\begin{cases} x + y = 10 \\ y + z = 14 \end{cases}$ ✗
- 3 eqⁿ 3 Unknowns $\begin{cases} x + y = 10 \\ y + z = 14 \\ x + z = 8 \end{cases}$ ✓

	②	③	④
Eg-1)	$+5x$	$-5x$	$+5x$
	$+3x$	$-3x$	$+3x$
	$8x$	$-8x$	$-2x$

- 5) $(+5) \times (+3) = 15$ $- \Rightarrow -$
- 6) $(-5) \times (-3) = +15$ $- - \Rightarrow +$
- 7) $(+5) \times (-3) = -15$ $- - - \Rightarrow -$
- 8) $(-5) \times (+3) = -15$ $- - - - \Rightarrow +$

- 1) $+5 + x = 10$ $x = 5$
- 2) $-5 + x = +0$ $x = -5$
- 3) $+2 - x = -7$ $x = 9$

$$4) \frac{x}{-5} = 10$$

$$x = -50$$

$$5) \frac{x}{+5} = 10$$

$$x = 50$$

$$6) \frac{-x}{2} = -20$$

$$-x = -40$$

$$x = 40$$

$$7) +5x = 10$$

$$x = 2$$

$$8) -5x = 10$$

$$x = -2$$

$$\text{Ex-1)} \quad 2x + 3y = 16 \quad \rightarrow \quad 2x + 3y = 16$$

$$3x + 4y = 7 \quad \rightarrow \quad -2x - 2y = -14$$

$$y = 2$$

$$x + 2 = 7$$

$$x = 5$$

• Home Work:

$$1) \quad x + y = 8$$

$$2x + 3y = 18$$

$$\rightarrow \quad 2x + 2y = 16$$

$$-2x - 3y = 18$$

$$-y = -2$$

$$y = 2$$

$$x + 2 = 8$$

$$x = 6$$

$$\therefore (6, 2)$$

$$2) \quad 3x + 2y = 22$$

$$x + y = 9$$

$$x + 13 = 9$$

$$x = -4$$

$$x + 5 = 9$$

$$x = 4$$

$$\rightarrow \quad 3x + 2y = 22$$

$$-3x - 3y = -27$$

$$-y = -5$$

$$y = 5$$

$$\therefore (4, 5)$$

$$3) \quad x - y = 4$$

$$3x + 4y = 10$$

$$x = \frac{26}{7}$$

$$26 - y = 4$$

$$26 - 4 = y$$

$$\frac{26-28}{7}$$

$$y = \frac{-2}{7}$$

$$\rightarrow \quad 4x - 4y = 16$$

$$3x + 4y = 10$$

$$x = 26$$

$$4) \quad -x + y = 12$$

$$2x - 3y = -31$$

$$\rightarrow \begin{array}{r} -2x + 2y = 24 \\ 2x - 3y = -31 \\ \hline -y = -7 \end{array}$$

$$\boxed{y = 7}$$

$$\therefore (-5, 7)$$

$$-x + 7 = 12$$

$$-x = 5$$

$$\boxed{x = -5}$$

$$5) \quad 4x + 5y = 41$$

$$-x - y = -9$$

$$\rightarrow \begin{array}{r} 4x + 5y = 41 \\ -4x - 4y = -36 \\ \hline y = 5 \end{array}$$

$$\therefore (4, 5)$$

$$-x - 5 = -9$$

$$-x = -4$$

$$\boxed{x = 4}$$

$$6) \quad 3x - 4y = 38$$

$$-x + y = -12$$

$$\begin{array}{r} 3x - 4y = 38 \\ -3x + 3y = -36 \\ \hline -y = 2 \end{array}$$

$$\boxed{y = -2}$$

$$\therefore (10, -2)$$

$$-x - 2 = -12$$

$$-x = -10$$

$$\boxed{x = 10}$$

$$7) \quad 5x + 2y = -28$$

$$-x + y = 14$$

$$\rightarrow \begin{array}{r} 5x + 2y = -28 \\ -5x + 5y = 70 \\ \hline 7y = 42 \end{array}$$

$$\boxed{y = 6}$$

$$\therefore (-8, 6)$$

$$-x + 6 = 14$$

$$-x = 8$$

$$\boxed{x = -8}$$

8)

$$-3x + 4y = 25$$

$$x + y = 1$$

$$-3x + 4y = 25$$

$$+3x + 3y = 3$$

$$7y = 28$$

$$y = 4$$

$$\therefore (-3, 4)$$

$$x + y = 1$$

$$x = -3$$

9)

$$7x + 3y = 13$$

$$-x - y = 1$$

$$7x + 3y = 13$$

$$-7x - 7y = 7$$

$$-4y = 20$$

$$y = \frac{20}{-4}$$

$$y = -5$$

$$\therefore (4, -5)$$

$$\begin{array}{l} -x = 1 - \frac{6}{4} \\ -x = \frac{-1}{2} \\ x = \frac{1}{2} \end{array}$$

$$-x = -4$$

$$x = 4$$

10)

$$-4x - 3y = 46$$

$$-x - y = 14$$

$$-4x - 3y = 46$$

$$4x + 4y = 56$$

$$y = -10$$

$$\therefore (-4, 10)$$

$$-x = 14$$

$$x = -14$$

11)

$$-x - 3y = 23$$

$$5x + y = -31$$

$$-5x - 15y = 115$$

$$5x + y = -31$$

$$-14y = 84$$

$$y = \frac{84}{-14}$$

$$y = -6$$

$$-x + 18 = 23$$

$$x = -5$$

$$\therefore (-5, -6)$$

$$12) \quad \begin{aligned} 4x - 9y &= 26 \\ -x - y &= -13 \end{aligned}$$

$$-x = -11$$

$$\cancel{x = 11} \quad \boxed{x = 11}$$

$$\begin{aligned} 4x - 9y &= 26 \\ -4x + 4y &= -52 \\ \hline -13y &= -26 \\ \boxed{y = 2} \end{aligned}$$

$$\therefore (11, -2)$$

$$13) \quad \begin{aligned} -6x + 5y &= 91 \\ -x + y &= 16 \end{aligned}$$

~~x~~

$$\begin{aligned} -6x + 5y &= 91 \\ 6x - 6y &= -96 \\ \hline -y &= -5 \\ \boxed{y = 5} \end{aligned}$$

$$\cancel{-x = 27}$$

$$-x = 11$$

$$\boxed{x = -11}$$

$$\therefore (-11, 5)$$

$$14) \quad \begin{aligned} 4x - y &= 24 \\ 3x + 2y &= -15 \end{aligned}$$

$$\begin{aligned} 8x - 2y &= 48 \\ 3x + 2y &= -15 \\ \hline 11x &= 33 \end{aligned}$$

$$12 - 24 = y$$

$$\boxed{y = -12}$$

$$\boxed{x = 3}$$

$$\therefore (3, -12)$$

$$15) \quad \begin{aligned} 7x + 3y &= -33 \\ x + y &= -7 \end{aligned}$$

$$\begin{aligned} 7x + 3y &= -33 \\ -7x - 7y &= 49 \\ \hline -4y &= 16 \end{aligned}$$

$$\boxed{x = -3}$$

$$\boxed{y = -4}$$

$$\therefore (-3, -4)$$

$$16) \quad \begin{aligned} -x - y &= -8 \\ 5x + 7y &= 44 \end{aligned}$$

$$\begin{aligned} -5x - 5y &= -40 \\ 5x + 7y &= 44 \\ \hline 2y &= 4 \\ \boxed{y} &= 2 \end{aligned}$$

$$\begin{aligned} -x &= 10 & -x &= -8 + 2 \\ \boxed{x} &= -10 & -x &= -6 \\ \boxed{x} &= 6 \end{aligned}$$

$$\therefore (-6, 2)$$

$$17) \quad \begin{aligned} 8x + 9y &= 13 \\ 6x + 7y &= 9 \end{aligned}$$

$$\begin{aligned} 8x + 9y &= 13 \\ 56x + 63y &= 91 \\ -54x + 63y &= 81 \\ \hline 2x &= 10 \\ \boxed{x} &= 5 \\ \therefore (5, -3) \end{aligned}$$

$$\begin{aligned} 30 + 7y &= 9 \\ 7y &= -21 \\ \boxed{y} &= -3 \end{aligned}$$

$$18) \quad \begin{aligned} 12x - 5y &= 14 \\ -8x + 9y &= -66 \end{aligned}$$

$$\begin{aligned} 108x - 45y &= 126 \\ -40x + 45y &= -330 \\ \hline 68x &= -204 \\ \boxed{x} &= -3 \end{aligned}$$

$$\begin{aligned} 24x + 9y &= -66 \\ 9y &= -90 \\ \boxed{y} &= -10 \end{aligned}$$

$$\therefore (-3, -10)$$

$$19) \quad \begin{aligned} 11x - 6y &= 106 & \times 2 \\ 17x - 4y &= 148 & \times 3 \end{aligned}$$

$$\begin{aligned} 22x - 12y &= 212 \\ -51x + 12y &= -144 \\ \hline -29x &= -232 \\ \boxed{x} &= 8 \end{aligned}$$

$$\begin{aligned} 88 - 6y &= 106 \\ -6y &= 18 \\ \boxed{y} &= -3 \end{aligned}$$

$$\therefore (8, -3)$$

$$20) \quad \begin{aligned} -13x + 5y &= -8 \\ 8x - y &= 7 \end{aligned}$$

$$-13x + 5y = -8$$

$$\underline{40x - 5y = 35}$$

$$27x = 27$$

$$\boxed{x=1}$$

$$-y = -1$$

$$\boxed{y=1}$$

$$\therefore (1, 1)$$

$$21) \quad \begin{aligned} 9x - 5y &= 194 \\ 4x + y &= 25 \end{aligned}$$

$$9x - 5y = 194$$

$$\underline{20x + 5y = 125}$$

$$29x = 319$$

$$\boxed{x=11}$$

$$y = 25 - 44$$

$$\boxed{y=-19}$$

$$\therefore (11, -19)$$

$$22) \quad \begin{aligned} -5x - 7y &= -8 \\ x + y &= 2 \end{aligned}$$

$$-5x - 7y = -8$$

$$\underline{7x + 7y = 14}$$

$$2x = 6$$

$$\boxed{x=3}$$

$$\boxed{y=-1}$$

$$\therefore (3, -1)$$

$$23) \quad \begin{aligned} 6x - 5y &= 81 \\ 7x + y &= 74 \end{aligned}$$

$$6x - 5y = 81$$

$$\underline{35x + 5y = 370}$$

$$41x = 451$$

$$\boxed{x=11}$$

$$-5y = 81 - 66$$

$$-5y = 15$$

$$\boxed{y=-3}$$

$$\therefore (11, -3)$$

24)

$$-4x + y = 27$$

$$6x - 5y = -65$$

$$-20x + 5y = 135$$

$$6x - 5y = -65$$

$$-14x = 70$$

$$\boxed{x = -5}$$

$$\boxed{y = 7}$$

$$\therefore (-5, 7)$$